

Ch 5: Elasticity and Its Application

How much quantity responds, and why it pays to know

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

From direction to size

Chapter 4 answered

Which **direction** does the price move?

Which way does quantity go?

Supply falls $\Rightarrow$ price up, quantity down. Every time.

Chapter 5 asks

By **how much**?

Why the size decides everything

A cigarette tax: mostly less smoking, or mostly more revenue?

A record harvest: good or bad news for farmers?

You raise your price 25%: do you earn more or less?

Same diagram, opposite answers, depending on **one number**

Monday: the Kerala paper, then measuring demand · Wednesday: revenue, supply, applications

The reading: Kerala, 1997-2001

The market

Sardine fishermen on the Kerala coast · fifteen beach markets

Markets open 5:00-8:00 AM · one market a day per boat · no storage, no ice

Over 70% of adults eat fish daily

The problem

At sea you cannot see any price.
You pick a beach before you know what it pays.

One morning: Tuesday, Jan 14, 1997

Market	Price (Rs/kg)
Badagara	0
Chombala	9.9
Quilandi	9.8

All within 15 km. Jensen (2007), Table I.

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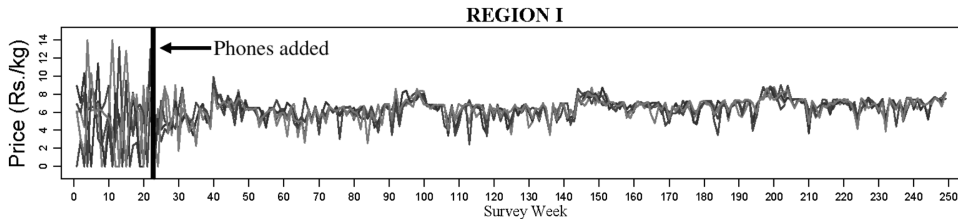
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Eleven boats at Badagara dumped the catch in the sea while buyers 15 km away went home empty — as much as 3,400 Rs in profit forgone per boat

The graph everyone remembers

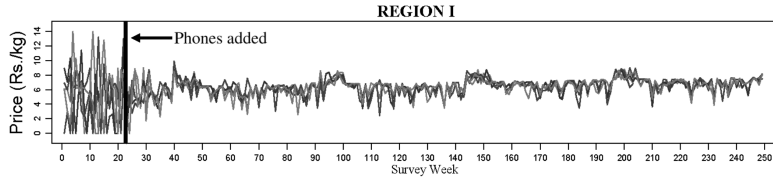


One line per beach market, Region I. 7:30–8:00 AM sardine price, 2001 Rs/kg. Jensen (2007), Fig. IV.

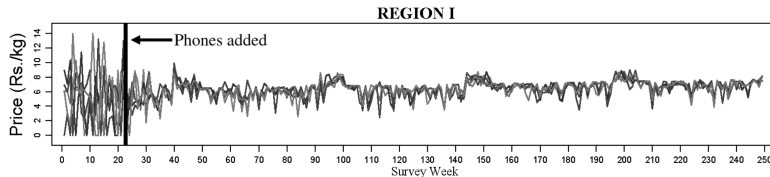
Question

- (a) One morning, one market pays 14 Rs/kg and another pays 0. How can fish be worth zero?
- (b) After week 23 the lines sit on top of each other. Who gained?

The graph everyone remembers

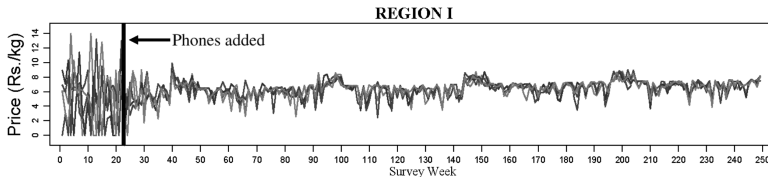


The graph everyone remembers



(a) A price of zero is a boat that reached a beach where the buyers had gone. The catch was **dumped**. Today's fish cannot wait, cannot travel, cannot be stored.

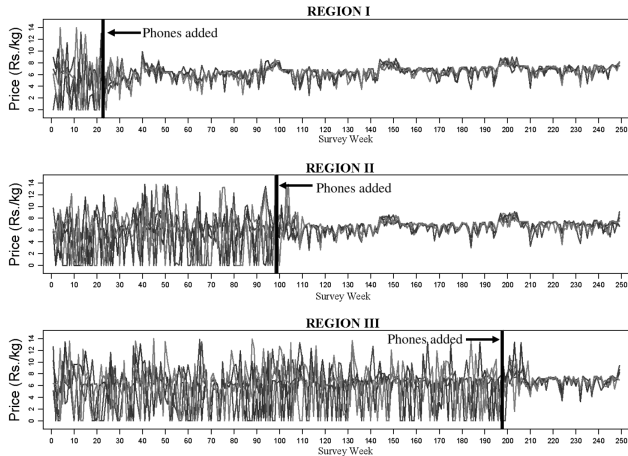
The graph everyone remembers



(a) A price of zero is a boat that reached a beach where the buyers had gone. The catch was **dumped**. Today's fish cannot wait, cannot travel, cannot be stored.

(b) The average hardly moves — the **spread** collapses. Fishermen earn 8% more, consumers pay 4% less, and the 5-8% of the catch that used to rot is now sold. Nobody caught an extra fish.

Three regions, three dates

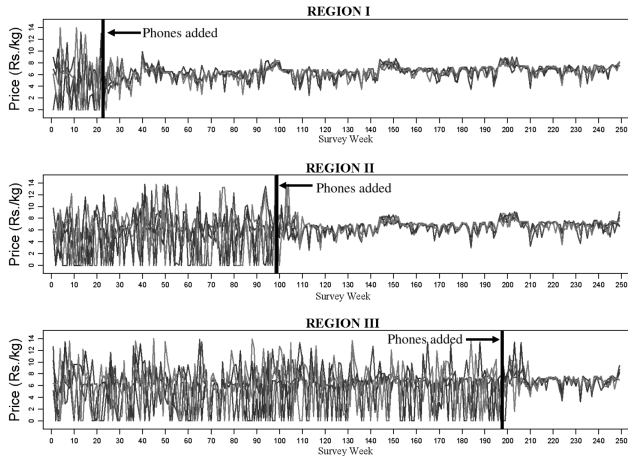


Phones reached region I at week 23, region II at week 98, region III at week 198.

Each series breaks at **its own** date, not at a common one.

Evidence, not coincidence: the same change, three times, whenever the phones arrived.

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A natural experiment: theory first, then data that could have refuted it.

What the phones did

Mobile service reached the coast **region by region**, 1997–2001. Fishermen called ahead and chose the beach.

	Before	After
Price dispersion	60–70%	$\leq 15\%$
One-price violations	50–60%	$\approx$ none
Catch wasted	5–8%	0
Fishermen's profits	—	+8%
Consumer price	—	–4%

Jensen, *QJE* 122(3), 2007.

Question

Profits up *and* prices down.
Where did the gain come from?

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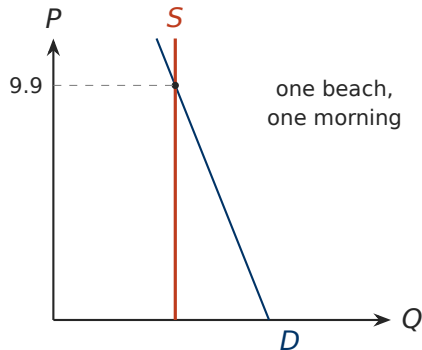
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Principle 6, measured: the signal must be **seen** to work.

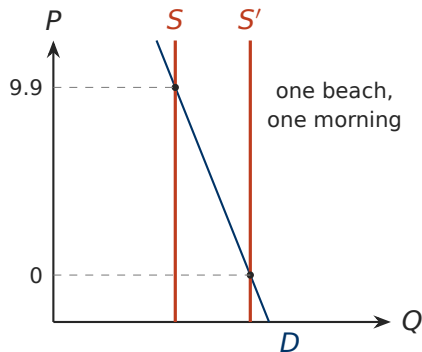
Why were the price swings so violent?



Today's catch is **already caught**: supply for the morning is a vertical line.

Local buyers are few, the market shuts at 8:00, the fish will not keep. Demand at one beach is **steep**.

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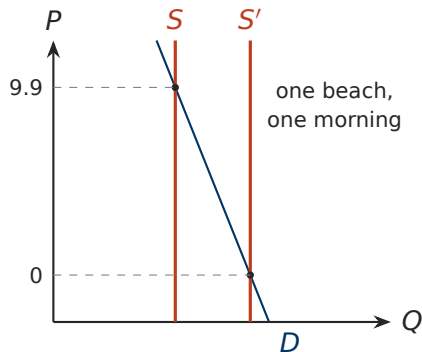


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The chapter in one line

How far the price moves depends on how **steep** — how inelastic — demand is. Elasticity measures that.

Your problem this semester

You build websites for local businesses.

You charge \$200 a site and sell 12 a month.

Your costs are rising — including the opportunity cost of your own time.

You are thinking about \$250.

Question

The law of demand says you will sell fewer. Fine.

How many fewer?

And does your revenue go up or down?

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“Demand slopes down” does not answer either question.

You need a number.

Price elasticity of demand

Elasticity

A measure of how much quantity demanded or supplied **responds** to a change in one of its determinants

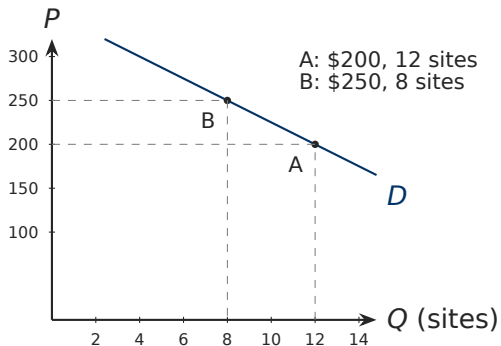
Price elasticity of demand

$$E_d = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}$$

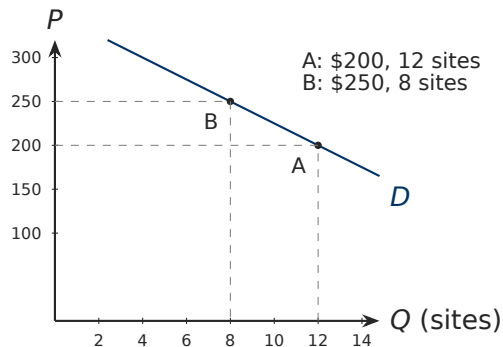
Along a demand curve P and Q move in **opposite** directions, so the ratio is negative. Convention: **drop the minus sign** and compare sizes.

Percentages, not units: rupees or dollars, kilos or tons, the number is the same.

First try: the obvious calculation



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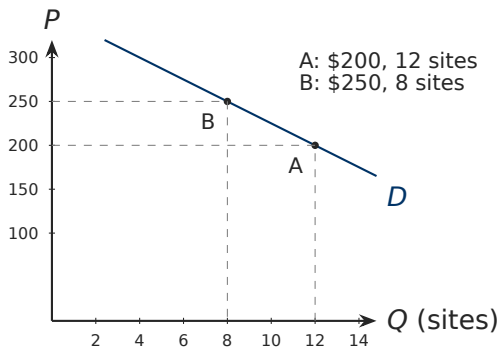
A $\rightarrow$ **B** (raise the price)

$$\% \Delta P = \frac{250 - 200}{200} = 25\%$$

$$\% \Delta Q = \frac{8 - 12}{12} = -33\%$$

$$E_d = 33/25 = 1.33$$

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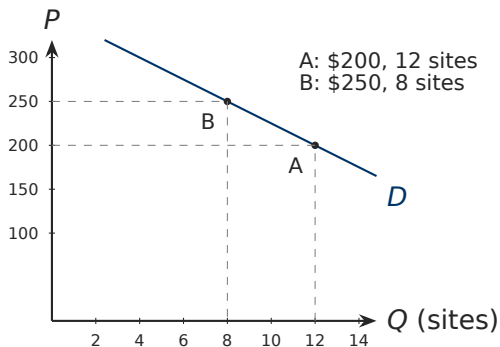
B → **A** (lower it again)

$$\% \Delta P = \frac{200 - 250}{250} = -20\%$$

$$\% \Delta Q = \frac{12 - 8}{8} = 50\%$$

$$E_d = 50/20 = 2.50$$

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Same two points, two answers. The base you divide by does the damage.

The midpoint method

Divide by the average, not by the starting point

$$\% \Delta Q = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \quad \% \Delta P = \frac{P_2 - P_1}{(P_1 + P_2)/2} \quad E_d = \left| \frac{\% \Delta Q}{\% \Delta P} \right|$$

The midpoint is the number halfway between the two values. Going up or going down now gives the **same** elasticity.

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\$250, 12 → 8 sites:

$$\% \Delta Q = \frac{-4}{10} = -40\% \quad \% \Delta P = \frac{50}{225} = 22.2\% \quad E_d = \frac{40}{22.2} = 1.8$$

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Quantity moves **almost twice** as hard as price. Hold on to that 1.8.

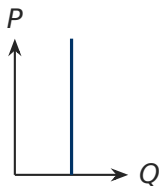
The vocabulary

Elasticity	Name	Meaning
$E_d = 0$	perfectly inelastic	quantity does not move at all
$E_d < 1$	inelastic	quantity moves <i>less</i> than price, in %
$E_d = 1$	unit elastic	they move together
$E_d > 1$	elastic	quantity moves <i>more</i> than price, in %
$E_d = \infty$	perfectly elastic	any price change wipes out the quantity

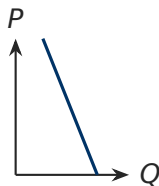
The dividing line is **1**: it is the line that decides what happens to revenue.

Same axes, same scale: the **flatter** the demand curve, the more elastic.

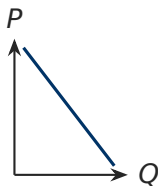
The five cases



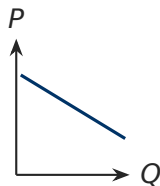
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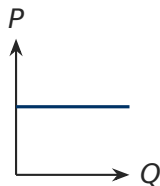
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Vertical: insulin for a diabetic this week · the sardines already in the boat

Horizontal: one farmer's wheat · one seller in a perfectly competitive market — charge a cent more and sell nothing

What makes demand elastic

Determinant	Elastic when	Example
Close substitutes	they exist	butter vs. margarine
Necessity or luxury	it is a luxury	sailboats, not insulin
Definition of the market	the market is narrow	Honda Civics vs. cars
Time horizon	the horizon is long	gasoline: a week vs. five years

Narrow beats broad. “Food” has no substitute; “vanilla ice cream” has forty.

Time is the quiet one. Gas doubles: this week you still drive to work; over five years you change car, job, home.

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A **brand** is a narrow market — which is why one seller’s demand is elastic when the industry’s is not.

Rank them

Question

Which has the more elastic demand, and why?

- (a) Insulin or vanilla ice cream
- (b) Gasoline over one week or gasoline over five years
- (c) Cigarettes or Marlboros
- (d) Fish at one beach at 7:45 AM or fish in Kerala over a month

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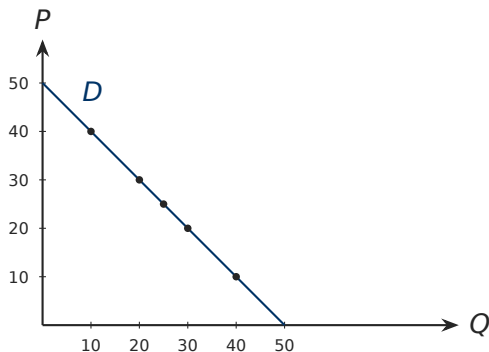
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- (d) The month. At 8:00 the unsold fish is worth nothing — hence 0 to 9.9.

Elasticity is not the slope

The Ch 4 curve, $Q_d = 50 - P$. One straight line, **constant slope**.

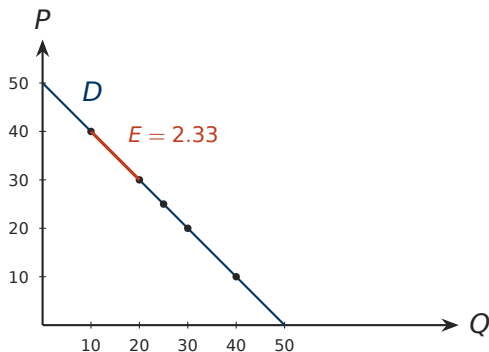
P	Q	$\% \Delta Q / \% \Delta P$	E_d
40 $\rightarrow$ 30	10 $\rightarrow$ 20	66.7/28.6	
30 $\rightarrow$ 20	20 $\rightarrow$ 30	40/40	
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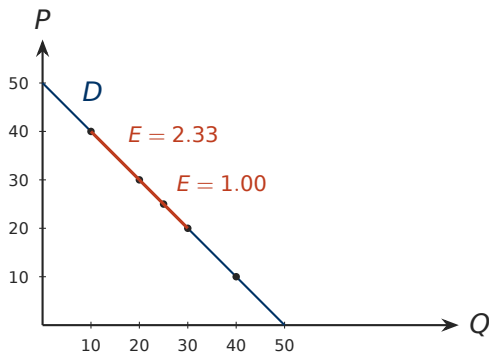
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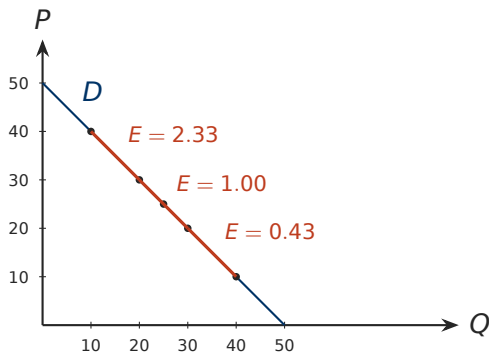
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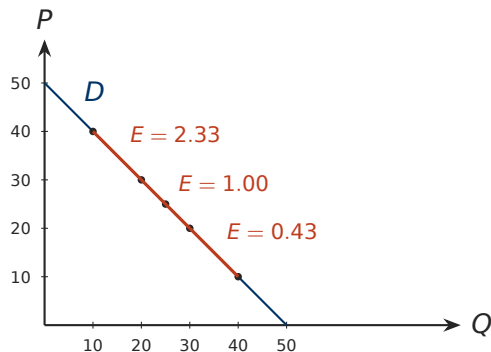
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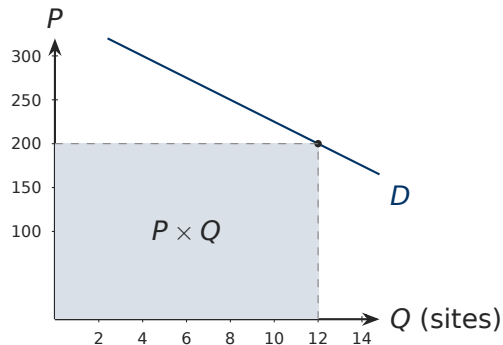
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High price: elastic. **Low price:** inelastic.

Elasticity **falls** as you move down the curve. Never say “the” elasticity of a linear curve.

Total revenue



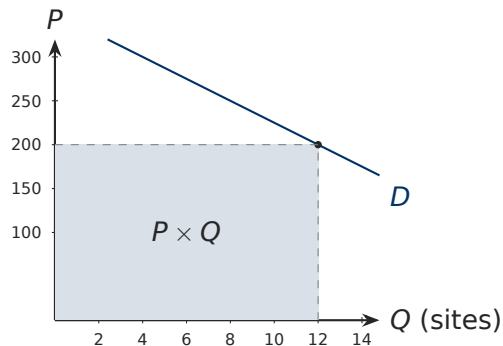
Total revenue

$TR = P \times Q$ — what sellers take in, what buyers spend. The **area** of the rectangle under the point.

Raise the price and two things happen:

- + every unit you still sell earns more
- you sell fewer units

Total revenue



Total revenue

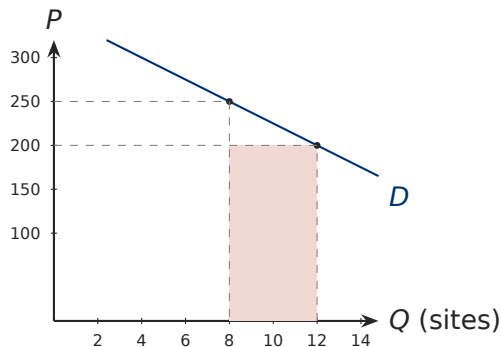
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The rectangle gets taller and narrower. Whether the **area** grows depends on which change is bigger in percent — on E_d .

Elastic: the price increase backfires

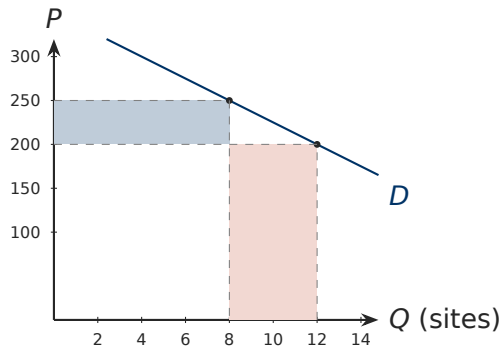


$E_d = 1.8$, elastic.

	P	Q	TR
Before	\$200	12	\$2,400
After	\$250	8	\$2,000

lost: fewer sites · gained: higher price

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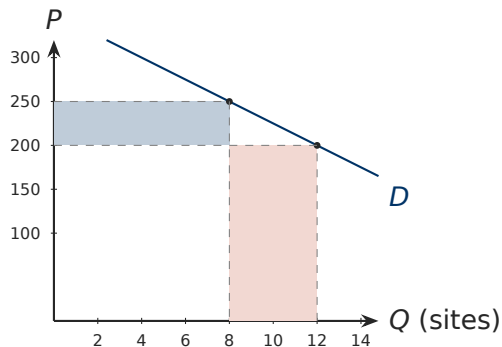


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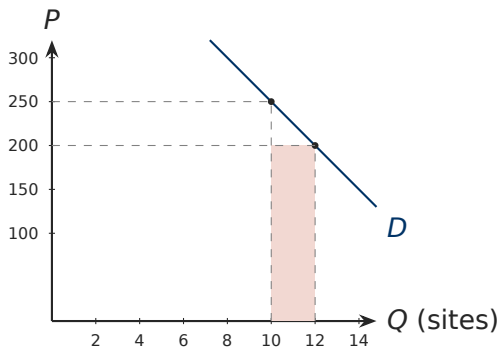
Quantity fell 40%, price rose 22%.
The loss is bigger.

When demand is elastic

Price up $\Rightarrow$ revenue **down**. Price
down $\Rightarrow$ revenue **up**.

Inelastic: the same increase pays

Same price rise, but now $E_d = 0.82$.

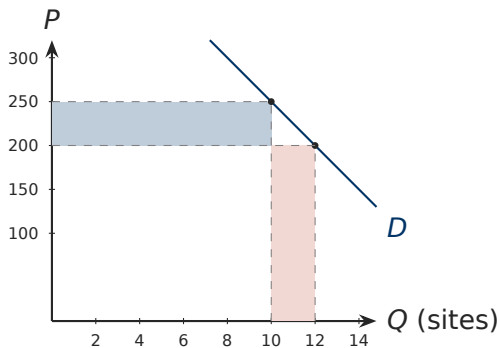


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a steeper curve: $12 \rightarrow 10$, not $12 \rightarrow 8$

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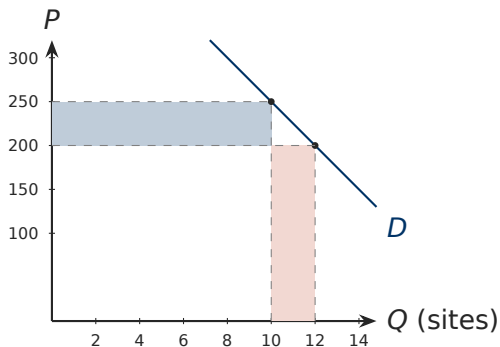
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Same price rise, but now $E_d = 0.82$.

	P	Q	TR
Before	\$200	12	\$2,400
After	\$250	10	\$2,500

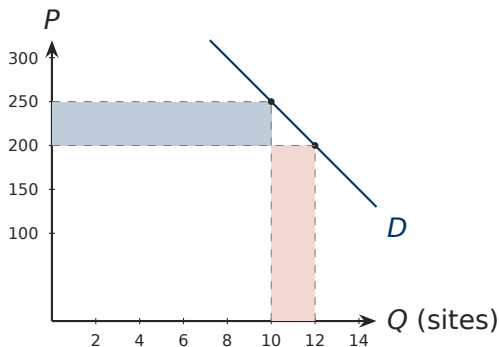
Quantity fell 18%, price rose 22%.
The gain is bigger.

When demand is inelastic

Price up $\Rightarrow$ revenue **up**. Price down
 $\Rightarrow$ revenue **down**.

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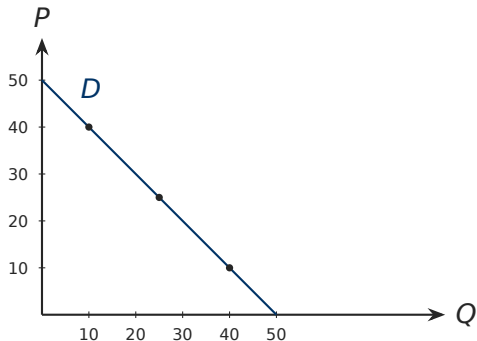
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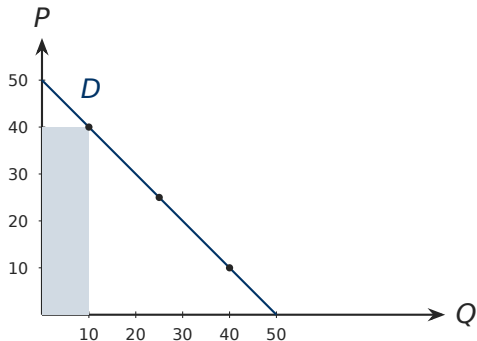
Nothing about the good changed. Only the **number**.

Revenue along the Ch 4 curve



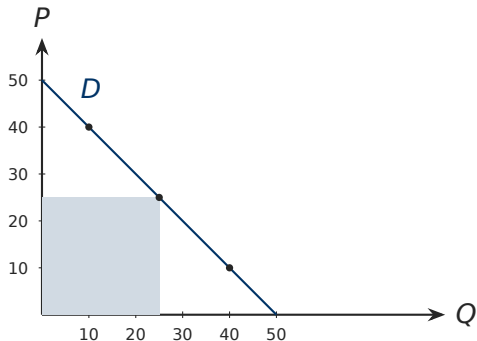
P	Q	TR
40	10	400
30	20	600
25	25	625
20	30	600
10	40	400

Revenue along the Ch 4 curve



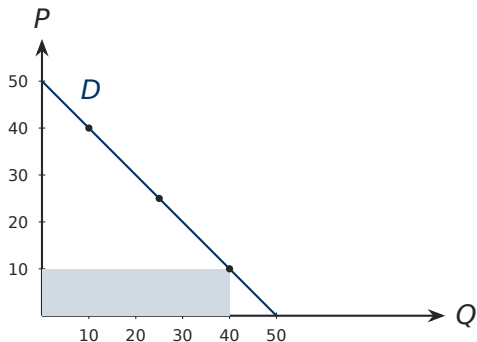
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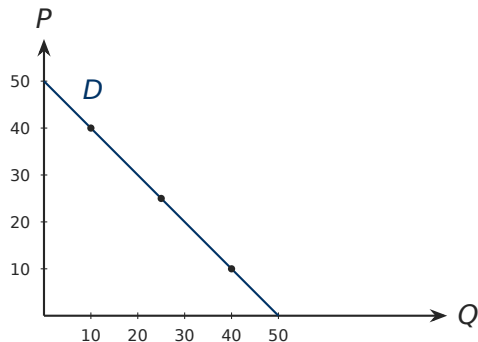
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Revenue peaks where $E_d = 1$, at the midpoint.

Above it, a price cut raises revenue;
below it, a price cut lowers revenue.

Question

- (a) Pharmacies raise the price of insulin by 10%. Does total spending on insulin rise or fall?
- (b) A fare war cuts the price of a luxury cruise by 20%. Does the cruise lines' revenue rise or fall?

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One rule, two different numbers, opposite answers.

Two more demand elasticities

Income elasticity

$$E_I = \frac{\% \Delta Q_d}{\% \Delta \text{income}}$$

Positive: normal good

Negative: inferior good (bus rides, instant noodles)

Among normal goods: necessities have small positive values (food, fuel), luxuries large ones (flights, jewelry)

Keep the sign here. It carries the information.

Cross-price elasticity

$$E_{xy} = \frac{\% \Delta Q_d \text{ of good } x}{\% \Delta P \text{ of good } y}$$

Positive: substitutes — Pepsi's price up, Coke's sales up

Negative: complements — printers dearer, ink sales down

Near zero: unrelated goods

This is the Ch 4 shifter table, now with a number attached.

Price elasticity of supply

Price elasticity of supply

$$E_s = \frac{\% \Delta Q_s}{\% \Delta P}$$

How much sellers respond to a better price. Same arithmetic, same midpoint method.

Inelastic supply

Beachfront land · a Rembrandt ·
this morning's catch, already
landed

No price makes more of it appear

Elastic supply

T-shirts, books, most
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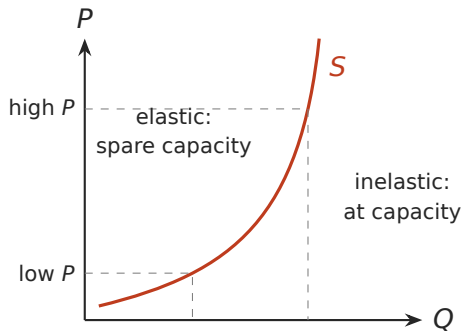
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Elastic supply

T-shirts, books, most
manufactured goods — firms with
spare capacity

Time again. Over a day, output is nearly fixed; over years, firms enter and build. **Supply is far more elastic in the long run.**

Supply near capacity

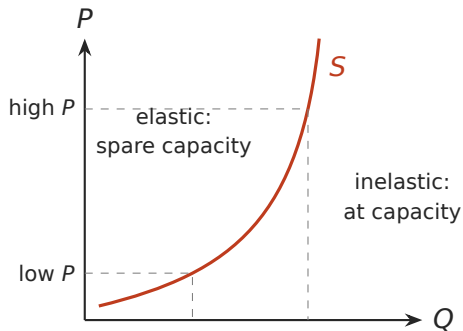


Supply elasticity is not one number either.

Low output: idle machines, unused hours. A small price rise brings out a lot more. **Elastic.**

Near capacity: plants running flat out. Even a large price rise brings out little. **Inelastic.**

Supply near capacity



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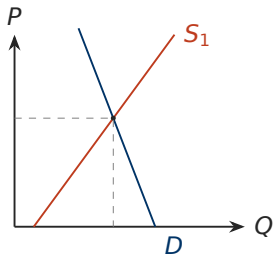
Near capacity: plants running flat out. Even a large price rise brings out little. **Inelastic**.

Same logic as the Kerala beach: when quantity cannot adjust, all the adjustment lands on the **price**.

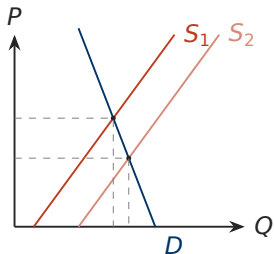
Application 1: good news for farming, bad news for farmers

Question

A new hybrid raises wheat yields 20%. Farmers adopt it. Are they better off?



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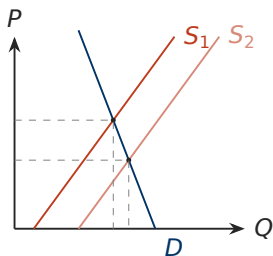


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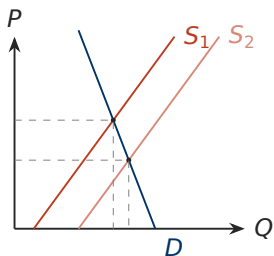
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Each farmer is rational; together they cut their incomes.

U.S. farm employment: 41% of the workforce in 1900, 1.9% in 2000.

Application 2: why OPEC could not hold the price up

Short run

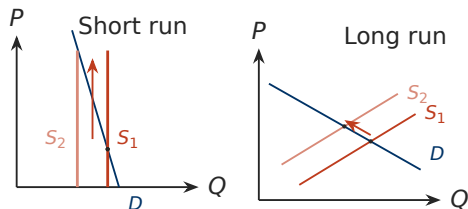
Supply (wells already drilled) and demand (your car, your furnace) are both **inelastic**

A cut in quantity $\Rightarrow$ a **huge** price jump

Long run

New fields open, cars get efficient — **both curves turn elastic**

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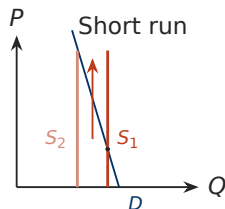


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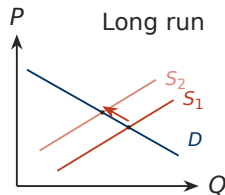
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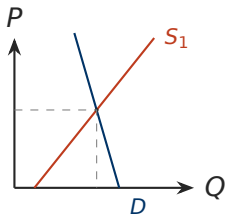
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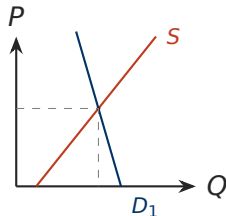
Crude, nominal: about \$3 a barrel before Oct 1973, \$11.65 posted in Jan 1974, over \$35 in 1980 — then a slide from \$26 to near \$10 during 1986.

Application 3: two ways to fight drug use

1. Interdiction

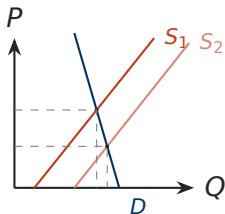


2. Treatment and education



Application 3: two ways to fight drug use

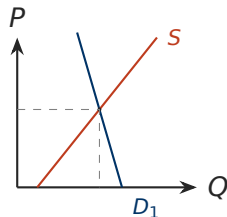
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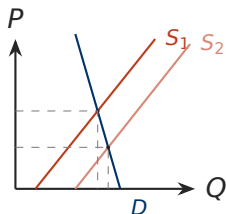
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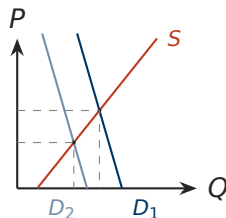
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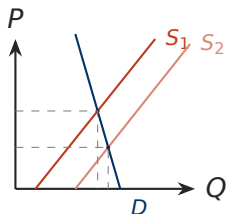
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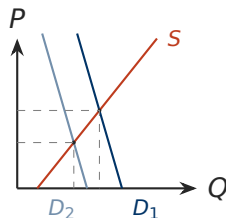
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Spending on drugs **falls**

Both cut the quantity consumed. Only one cuts the **sellers' revenue** — what finances the trade, and what users steal to pay.

The argument turns on demand being inelastic. If it were elastic, interdiction would cut seller revenue too.

Exam-style: cigarettes

Question

The price elasticity of demand for cigarettes is about 0.2. A pack costs \$5.

- (a) What price would cut smoking by 10%?
- (b) Is such a tax more effective in the short run or the long run?
- (c) Are teenagers more or less price-sensitive than adults?

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(c) More sensitive: less income, and not yet addicted. Addiction is what makes adult demand inelastic.

Summary

Measuring

- Elasticity = % change in quantity over % change in price; report it positive
- Midpoint method: divide by the average, so up and down agree
- Elastic > 1 , inelastic < 1 , unit = 1
- Elastic when there are substitutes, when it is a luxury, when the market is narrow, when there is time
- A linear demand curve has every elasticity along it

Using

- $TR = P \times Q$; elastic $\Rightarrow$ price and revenue move *opposite*, inelastic $\Rightarrow$ *together*
- Revenue peaks where $E_d = 1$
- Income elasticity signs normal vs. inferior; cross-price signs substitutes vs. complements
- Supply: inelastic when quantity is fixed or capacity is full, elastic in the long run
- Whichever side cannot adjust, the **price** does the adjusting

Next: Ch 6, Supply, Demand, and Government Policies.

MindTap: Ch 5 homework due Sunday, 11 PM.